

Solution Requirements

Date	30 September 2026
Team ID	SWTID-2026-1418
Project Name	Fit buddy
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely register, log in, and log out using email and password. User credentials must be securely stored</i>	High
2	Authorization levels	<i>The system provides user and administrator roles. Users can manage their own fitness data, while administrators can manage users and system information.</i>	Medium

3	External interfaces	<i>The application connects with an AI/LLM API to provide personalized fitness and nutrition recommendations.</i> <i>It may also connect with databases and fitness-related APIs.</i>	High
4	Transactions processing	<i>The system processes user inputs such as fitness goals, body details, workout data, and food information, and generates personalized recommendations based on the submitted data.</i>	High
5	Reporting	<i>Users can view fitness progress, workout history, nutrition information, goals, and personalized recommendations through dashboards and reports.</i>	High
6	Business rules, etc.	<i>The system generates recommendations based on user goals, fitness level, preferences, available equipment, and other entered information. Recommendations should be personalized for each user.</i>	High
7	Compliance to laws or regulations	<i>The system must protect personal information and follow applicable data privacy and security practices. Users should be informed about how their data is collected and used.</i>	High

8	Other	<i>The system provides an AI-powered fitness assistant that can answer fitness-related questions and provide workout and nutrition guidance.</i>	High
---	-------	--	------

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The application should respond quickly to user actions and display dashboards and recommendations without unnecessary delay.</i>	<i>Normal pages should load within 2 seconds; API responses within 3 seconds under normal load.</i>
2	Scalability	<i>The system should support increasing numbers of users, fitness records, and AI requests without major performance degradation.</i>	<i>Supports at least 1,000 concurrent users during initial deployment.</i>
3	Security & Data Privacy	<i>User passwords and personal fitness data must be protected using secure authentication, encryption, access control, and secure API communication.</i>	<i>HTTPS/TLS for data transmission; passwords securely hashed; unauthorized users cannot access another user's data.)</i>
4	Reliability & Availability	<i>The application should remain available and handle errors gracefully without losing user data.</i>	<i>Target 99% monthly availability and proper error handling for failed API/database requests.</i>

5	Usability & Accessibility	<i>The interface should be simple, responsive, mobile-friendly, and easy to use for users with different levels of technical knowledge.</i>	<i>Responsive on desktop, tablet, and mobile; clear navigation and readable interface.</i>
6	<i>Other</i>	<i>The application should be maintainable, modular, portable, and easy to update. Backend, frontend, database, and AI components should be organized separately.</i>	Code should be modular and documented; application should run consistently in the supported development environment.